

GraphOne / FrontierAtlas — System Architecture Document

Production-Grade Data Pipeline & LLM Intelligence Orchestration Engine

1. Massive Scale Strategy (500,000+ Entities Ingestion)

To scale entity ingestion across hundreds of thousands of records without operational bottlenecks, the pipeline decouples link discovery from page processing using a distributed asynchronous architecture.

- **Distributed Worker Pools:** Async event loops using Python asyncio and aiohttp paired with dynamic worker pools (Celery/Redis) scale horizontally without changing core logic.
- **Anti-Bot Evasion Engine:** High-value targets (Cloudflare/DataDome protected) are managed via Playwright Async running stealth drivers and dynamic proxy rotation (BrightData/ScraperAPI).
- **Concurrency Throttling:** Semaphore-bounded request queues maintain optimal target throughput without causing IP bans[cite: 1].

2. Resilient Multi-Tier LLM Engine (413 & 429 Mitigation)

Large Language Model calls are managed through a robust execution layer designed for max data density and automated fault handling.

- **DOM Stripping & Content Truncation (HTTP 413 Prevention):** Raw HTML payloads are pre-processed using trafilatura/BeautifulSoup to remove scripts, styles, headers, and navigation footers, maximizing token efficiency[cite: 1].
- **Multi-Tier Fallback Cascade:** API calls systematically cascade across model tiers: **Gemini Flash** (Primary) → **Groq Llama 3** (Secondary) → **DeepSeek** (Tertiary)[cite: 1].
- **Exponential Backoff with Jitter (HTTP 429 Rate Limits):** Rate-limit exceptions trigger dynamic sleep backoffs calculated as: $Wait\ Time = 2^{retry_count} + uniform(0, 1)$ [cite: 1].

3. Freshness Tracking & Entity Resolution Strategy

Ensures absolute data accuracy and strict 24-hour publication freshness across news and job market channels.

- **Distributed Deduplication:** SHA-256 content/URL hashing stored in a Redis Bloom Filter guarantees O(1) duplicate checks across crawling instances[cite: 1].
- **Timestamp Normalization:** Dynamic time strings (e.g., '2 hours ago', meta tags) are parsed into standardized ISO-8601 UTC format, automatically discarding records older than 24 hours[cite: 1].
- **Deterministic Entity Resolution:** Extracted strings undergo legal suffix stripping, normalization, and fuzzy matching (Token Sort Ratio ≥ 85%) against a canonical seed database[cite: 1].

4. System Storage & Database Architecture Choice

The storage architecture divides operational transactions from analytical graph queries to maximize throughput.

Storage Layer	Technology	Architectural Justification
Relational DB	PostgreSQL	Stores schema-validated JSONB entity records (Startups, Products, Papers, Jobs)[cite: 1].
Graph Database	Neo4j	Maps multi-dimensional relations (Founders → Startups → Products → Papers)[cite: 1].
Vector Store	Pinecone / Qdrant	Enables semantic vector search over paper abstracts and startup profiles.

In-Memory Cache	Redis	Manages distributed task queuing, rate limit counters, and deduplication states[cite: 1].
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